

Completing the square



- 1 Explain how to find the number to add to $x^2 - 5x$ to make it a perfect square trinomial.
- 2 Consider the expression $x^2 + 6x$.
 - a Does adding 36 to the expression make it a perfect square?
 - b What should be added to $x^2 + 6x$ to make it a perfect square?
- 3 Consider the equation $x^2 - 3x = 6$. What should be added to the equation in order to complete the square?
- 4 Complete the following expressions so they form a perfect square:
 - a $x^2 + 10x + \square$
 - b $x^2 - 2x + \square$
 - c $x^2 - 11x + 16$
 - d $x^2 - x + \square$
 - e $x^2 + mx + \square$
 - f $x^2 - \frac{4}{5}x + \square$
- 5 Complete the following perfect squares:
 - a $x^2 + 4x + \square = (x + \square)^2$
 - b $x^2 - 5x + \square = (x - \square)^2$
 - c $x^2 - \frac{7}{4}x + \square = (x - \square)^2$
- 6 Rewrite the following in the form $(x + b)^2 + c$:
 - a $x^2 + 12x$
 - b $x^2 - 12x$
 - c $x^2 + 2x + 2$
 - d $x^2 - 14x + 56$
 - e $x^2 + 4x + 2$
 - f $x^2 - 6x + 5$
 - g $x^2 + 3x + 6$
 - h $x^2 - 7x + 14$
- 7 Rewrite the following in the form $a(x + b)^2 + c$:
 - a $3x^2 + 9x + 8$
 - b $4x^2 - 11x + 7$
- 8 Rewrite $4x^2 - 36x + 79$ in the form $c((x + a)^2 + b)$.

9 Factor the following:



a $b^2 + 14b + 49$

c $x^2 + 10x + 25$

e $x^2 - 12x + 36$

g $81x^2 - 72x + 16$

i $49x^2 - 28x + 4$

k $36 - 12x + x^2$

m $7d^2 + 112d + 448$

o $3x^2 - 24xy + 48y^2$

q $-3x^2 + 12x - 12$

s $x^2 + 3x + \frac{9}{4}$

b $y^2 - 10y + 25$

d $x^2 + 14x + 49$

f $x^2 - 16x + 64$

h $81x^2 + 36x + 4$

j $81 + 18x + x^2$

l $25 + 60r + 36r^2$

n $4x^2 + 40x + 100$

p $-6r^2 - 96r - 384$

r $b^2 - 7b + \frac{49}{4}$

t $x^2 - 3x + \frac{9}{4}$

10 Factor the quadratics using the method of completing the square to get them into the form:

$$y = (x - a + \sqrt{b})(x - a - \sqrt{b})$$

a $y = x^2 - 6x + 7$

b $y = x^2 + 8x + 14$